Data Governance, Compliance and Ethics

SUPPLEMENTARY MATERIAL (MODELS AND TOOLS FOR DG, RISK MANAGEMENT)

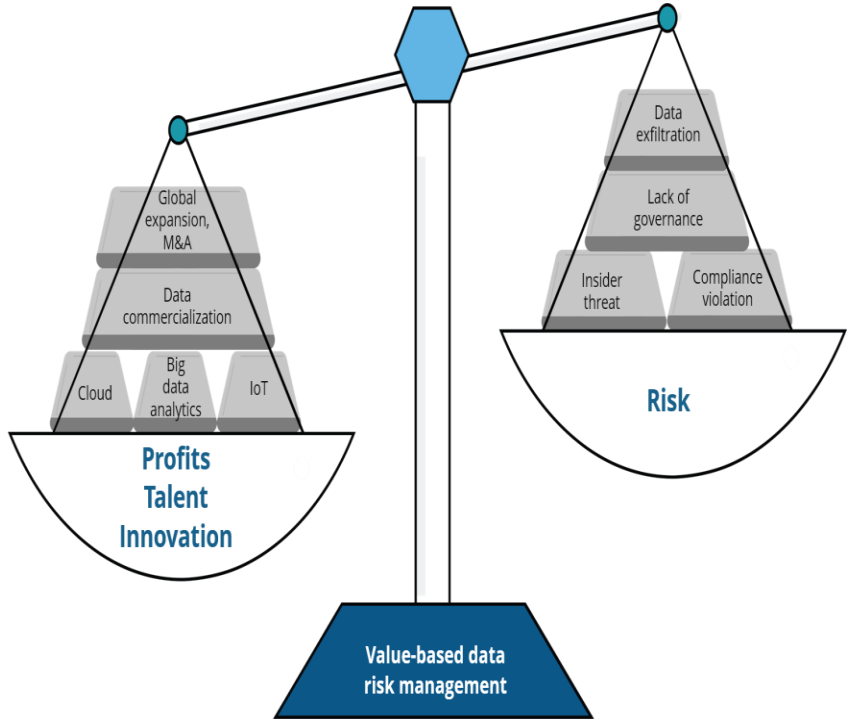


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What is Data Risk Management?

Data risk is the potential for business loss due to poor **data** governance.

Data governance and Risk management is fundamental to risk mitigation and management and therefore crucial for any data-driven business.



"There are only two types of companies: those that have been hacked and those that will be." Former FBI Director Robert Mueller (2012)



How Does Data Governance Help an Organization Manage Information Risk?

Resources, people and finances to implement data security measures are limited

Organisations need to know the nature, extent and probabilities of the risk that face them so that they can allocate appropriate resources to protecting that data for particularly

- the most sensitive and critical data to your organization
- the most valuable information

So how do you identify and quantify risk?



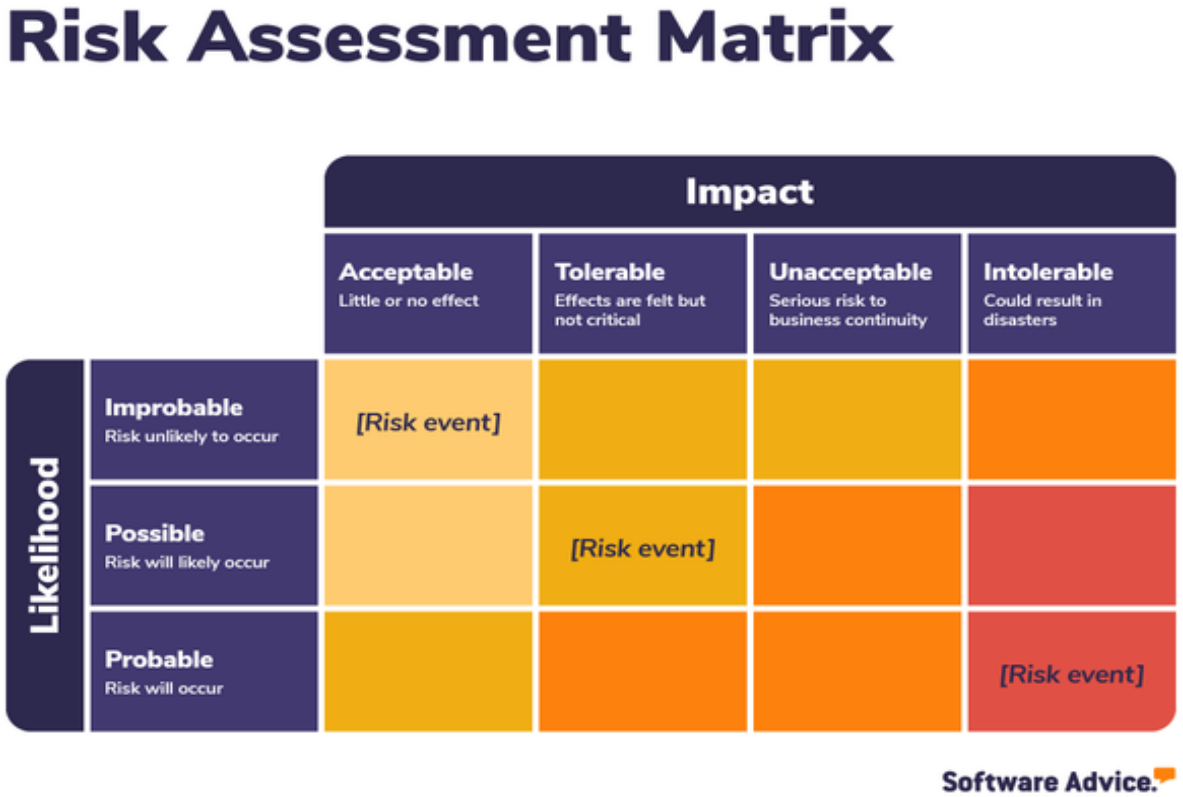
Organisations should take a risk-based approach and focus on the most sensitive data assets.

Identifying and Assessing Data Risk

Identification (structured walkthroughs, external assessment ("ethical hackers"), benchmarking, monitoring tools, audits)

Quantification (probability of negative event, cost of event) and rank risks

Decide on appropriate cut-off for mandating action



Matching risk scenarios with actions

Identify high level risk scenario causing potentially risk of financial loss or damage and match with appropriate level/action under data governance strategy management of data. Some examples of this approach are below:

High level Scenario	Related aspects of Data Management
Poor management of data	Data Governance – roles/responsibilities
Breach of data protection laws	Information security & Privacy
Data decay of data required by compliance	Information lifecycle management
Data integration errors	Data/Information Architecture
Inability to perform regulatory / statutory reporting	Data Quality – Timeliness
Inaccurate, untimely or poor quality management	Data Quality assessment and management
Multiple data sources / lack of definitive data sources for key data	Metadata management/ Sourcing
Reliance on third parties for base data	Third party sourcing
Inadequate activity management/segregation of roles	Data Governance
lack of management review	Data Governance

Models and tools for data governance



Data governance tools



Data governance is a set of policies, procedures, protocols, and metrics that control how data is used, managed, and stored.

Many organizations struggle with implementing an effective data governance approach in the face of so many dependencies. To ease this resource-heavy process, more businesses are adopting solutions to help organize, access, and protect data on a daily basis. These solutions, or data governance tools, help automate the workflow and provide structure to large volumes of data.

Any data governance tool must be able to quickly and effectively manage data from many different access or storage points as well as meet the needs of different end-users.

Gartner Magic Quadrant for data quality solutions

Gartner compares

- Connectivity
- Data profiling, measurement, and visualization
- Monitoring
- Parsing:
- Standardization and cleaning:
- Matching, linking and merging

And a range of other features



Popular Data Governance Tools (1/2)

Data Governance Tools	Features & Functions	Platform	Price
OvalEdge 	Data Governance, Data Catalog, Automated Data Lineage, Data Discovery, Self-Service Analytics	Windows, Unix, Cloud, On-Premise, Web, and SaaS.	Starts at $50/user/month. For more details, contact company.
Truedat 	Data Governance, Business Glossary, Data Catalog, Data Lineage, Data Quality.	Cloud, On-Premise.	Open source. Professional Services Fee. Contact the company, for more details.
Collibra 	Collaboration Features, Data Help Desk, Automation of Data Governance & Management.	Windows, Mac, iOS, Cloud, Web, and SaaS.	Contact company.

Popular Data Governance Tools (2/2)

Data Governance Tools	Features & Functions	Platform	Price
IBM 	Data Governance, Data Cataloging, Obtaining Information for Big Data Projects.	Windows, Cloud, Web, and SaaS.	Contact company.
Talend 	Data Governance, Cloud Integration, Data Integration, API, & Application Integration.	Windows, Mac, Cloud, Web, and SaaS.	Talend Open Source: Free Stich Data Loader: $100-$1000. Talend Cloud Data Integration: $1170/user/ month.
Informatica 	Manage GDPR Data Risks, Detect & Protect Sensitive Customer Data, Verify Contact Data.	--	Starts at $2000/month.
Alteryx 	Discover, Prepare, & Analyze the Data. Deploy & Share Analytics. Collaboration Features.	Windows, Mac, Cloud, Web, and SaaS.	Alteryx Designer: Starts at $5195/user /year. Alteryx Server: Starts at $58,500.

Recap

DATA GOVERNANCE includes the **people, processes and technologies** needed to manage and protect the company's data assets in order to guarantee generally understandable, correct, complete, trustworthy, secure and discoverable corporate data.

DATA MANAGEMENT is the process of ingesting, storing, organizing and maintaining the data created and collected by an organization.

DATA QUALITY – ensuring data is correct, consistent and free of "noise" that might impede usage and analysis.

DATA PROVENANCE is the documentation of where a piece of data comes from and the processes and methodology by which it was produced.

Questions?

